

Quiz 1

● Graded

Student

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Total Points

10 / 10 pts

Question 1

Neural LM

2 / 2 pts

✓ + 2 pts Correct

+ 0 pts Incorrect

Question 2

(no title)

4 / 4 pts

2.1 PMI

3 / 3 pts

✓ + 3 pts Correct

+ 0 pts Incorrect

2.2 (no title)

1 / 1 pt

✓ + 1 pt Correct

+ 0 pts Incorrect

Question 3

One-Hot Vector

2 / 2 pts

✓ + 2 pts Correct

+ 0 pts Incorrect

Question 4

(no title)

2 / 2 pts

✓ + 2 pts Correct

+ 0 pts Incorrect

Q1 Neural LM

2 Points

Which of the following statement about Feedforward Neural LMs is true?

- ☐ Feedforward LMs are generally more expressive than RNN LMs.
- ☐ Feedforward LMs can handle variable input context lengths.
- ☐ Feedforward LMs are good at solving sequence-to-sequence tasks, such as translation or summarization.
- ☒ Feedforward language models predict the next word using a fixed-size context window, with the entire prediction made by a feedforward neural network

Q2**4 Points**

In class, we learned that pointwise information (PMI) measures whether words x and y co-occur more than if they were independent. Below is a matrix with 5 rows (words) and 5 columns (context). Please answer the following questions.

	bike	mascot	bus	lunch	physics
unicycle	4	1	1	0	0
school	1	2	1	1	2
hoverboard	2	1	3	0	0
cafeteria	0	1	0	3	1
math	0	1	0	0	6

Note: You can use a calculator for this question.

Q2.1 PMI**3 Points**

What is $P(w = \text{school})$, $P(c = \text{physics})$ and $P(w = \text{school}, c = \text{physics})$?

Round your answer to **two decimal places even with zeros**. For example, if the answer is 1, put 1.00 into the text box.

$P(w = \text{school})$

0.23

$P(c = \text{physics})$

0.29

$P(w = \text{school}, c = \text{physics})$

0.06

Q2.2

1 Point

Remember

$$PMI(x, y) = \log_2 \frac{p(x, y)}{p(x)p(y)}$$

Using the results in Q2.1 to calculate the value of PMI (physics,school):

Round your answer to **two decimal places even with zeros**. For example, if the answer is 1, put 1.00 into the text box.

You can us

PMI (physics, school)

-0.02

Q3 One-Hot Vector

2 Points

When we're trying to represent words in discrete symbols, one possible way is to do one-hot encoding. However, these naive methods have some issues that limit their applicability. Which of the following choices is not the issue of one-hot encoding?

- ☐ The dimensionality of the vector is large.
- ☐ It cannot represent new (out-of-vocabulary) words.
- ☐ It cannot capture similarity between words.
- ☒ It cannot represent infrequent words in the training set.

None of the above.

Q4**2 Points**

Assume we are learning d -dimensional word embeddings using the Word2Vec skip-gram algorithm, and the training corpus contains $|V|$ unique words. How many parameters does the model need to learn?

- ☒ $2|V|d$
- ☐ $|V|d$
- ☐ $3|V|d$
- ☐ $2d^2$